# EEG CLASSIFICATION PROJECT

## EEG-BASED CLASSIFICATION OF AWAKE AND PROPOFOL SEDATED STATES USING INFORMATION-THEORETIC AND SPECTRAL FEATURES

# OBJECTIVES

- Are IIT proxies sensitive to changes in consciousness?
- Are IIT proxies as informative as conventional spectral features?
- Do IIT proxies provide complementary information beyond spectral power?, Which IIT proxy is most useful?
- Does combining them improve performance?

# BACKGROUND INFORMATION

- EEG signals are typically decomposed into different frequency bands, each associated with distinct neural processes:
  - Delta band: predominates during deep sleep;
  - Theta band: involved in memory encoding and retrieval, and associated with power increases during drowsiness;
  - Alpha band: involved in motor functions (over the motor cortex) and cognitive functioning;
  - Beta band: indicator of cortical transmission, and associated with power increases during active concentration.

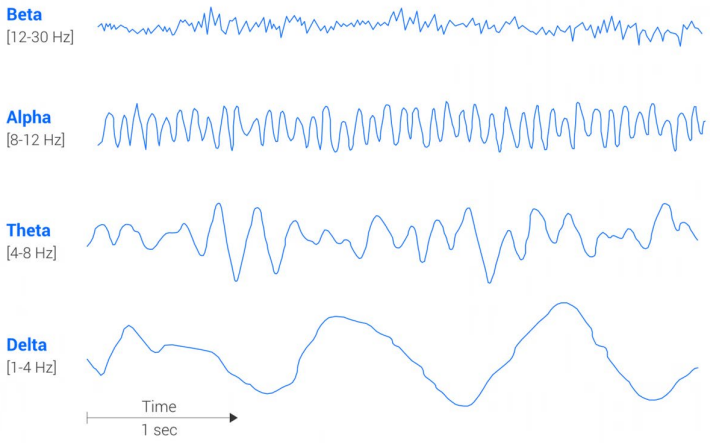



**Figure 1.** EEG brain wave patterns for the four frequency bands (delta, theta, alpha, beta).

https://www.bitbrain.com/blog/qeeg-brain-mapping

# BACKGROUND INFORMATION

- **Integrated information theory (IIT)** proposes a mathematical model for the consciousness of a system. It comprises a framework ultimately intended to explain why some physical systems (such as human brains) are conscious, and to be capable of providing a concrete inference about whether any physical system is conscious, to what degree, and what particular experience it has; why they feel the particular way they do in particular states and what it would take for other physical systems to be conscious .The theory inspired the development of new clinical techniques to empirically assess consciousness in unresponsive patients. According to IIT, **integrated information (Φ)** corresponds to the quantity of consciousness. That is, a system's consciousness (what it is like subjectively) is conjectured to be mathematically described by the system's causal structure (what it is like objectively). Therefore, it should be possible to account for the conscious experience of a physical system by unfolding its complete causal powers.

# METHODOLOGY

- OpenNeuro ds005620 [A repeated awakening study exploring the capacity of complexity measures to capture dreaming during propofol sedation]
- Data :
  - Uses dataset from openNeuro(ds005620)
  - Uses eeg data from awake (eye close) and propofol-sedated subjects
- Preprocessing
  - Average re-referencing
  - High and low pass filtering
- Epoching
  - eeg signal from each subject were sliced into windows of size 50000 s and step size 100. Because of limitation in computation resources, only 100 windows per subject are selected to compute feature values for the final datasets

# DATASET

| | subject | alpha | beta | theta | delta | o-info | MI | complexity | label |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 1010 | -4.48E-08 | -3.10E-08 | 3.12E-08 | 1.27E-06 | 0.309158 | 1.499439 | 0.040959 | awake |
| 1 | 1010 | 9.34E-09 | 7.27E-10 | 2.60E-09 | 1.25E-06 | 0.286655 | 1.481759 | 0.042597 | awake |
| 2 | 1010 | 5.32E-08 | 1.38E-08 | -4.01E-08 | 1.23E-06 | 0.278759 | 1.473252 | 0.044236 | awake |
| 3 | 1010 | 2.17E-08 | 2.10E-09 | -8.60E-08 | 1.21E-06 | 0.267963 | 1.442224 | 0.044645 | awake |
| 4 | 1010 | -4.81E-08 | -9.28E-09 | -1.07E-07 | 1.21E-06 | 0.263767 | 1.447043 | 0.045055 | awake |
| 5 | 1010 | -6.38E-08 | -7.77E-09 | -8.04E-08 | 1.22E-06 | 0.343838 | 1.462391 | 0.043417 | awake |
| 6 | 1010 | -4.65E-09 | -1.06E-09 | -1.42E-08 | 1.23E-06 | 0.431755 | 1.499119 | 0.043417 | awake |

Channels:

- data from 6 channels were used to compute features
- Channels: Fz,Cz,Pz,P3,P4,Oz were selected because they provide broad coverage of the brain's major cortical regions

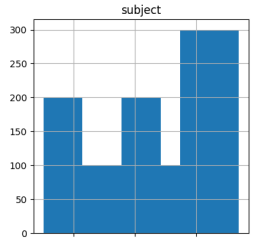
- Features:
  - Spectral features: alpha, beta, theta, delta waves
  - IIT proxy: O-information, Mean pairwise mutual information , Lempel–Ziv complexity
- Label:
  - awake=1 (2000 samples)
  - sedated = 0 (2100 samples)

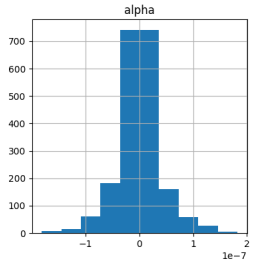
# IIT proxy

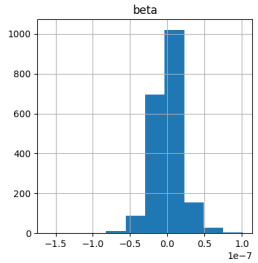
- Since direct computation of integrated information (Φ) is computationally difficult  for EEG, this project evaluated information-theoretic proxy measures instead.
- O-information
  - O-information measures the difference between the total correlation and the dual total correlation and it is shown to be a proxy of the difference between redundancy and synergy: when the O-information of a set of variables is positive this indicates redundancy, when it is negative, synergy.
- Mean pairwise mutual information
  - In probability theory and information theory, the mutual information of two random variables is a measure of the mutual dependence between the two variables. More specifically, it quantifies the amount of information obtained about one random variable by observing the other random variable.  Hence, MI feature in this project is obtained from calculating the mean of mutual information of every possible pair of eeg channels.
- Lempel–Ziv complexity
  - Lempel-Ziv (LZ) complexity is a mathematical metric that quantifies how many distinct, non-overlapping patterns exist within a sequence of data. Originally developed for lossless data compression, it is widely used today to measure the unpredictability and structural richness of time-series data, such as brainwaves, heartbeats, and machine vibrations

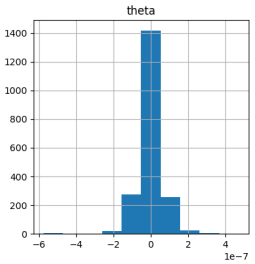
# ANALYSIS

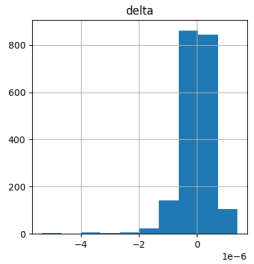
## awake

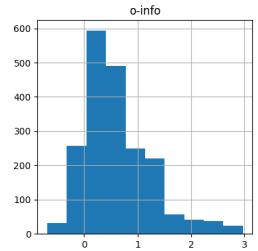


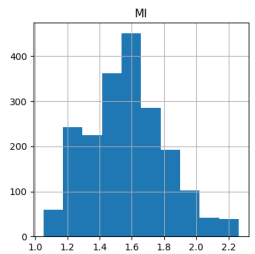



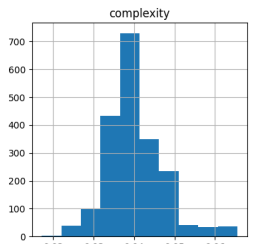



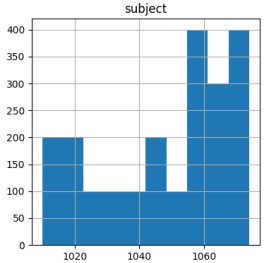



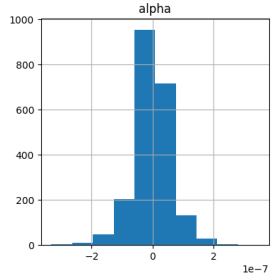



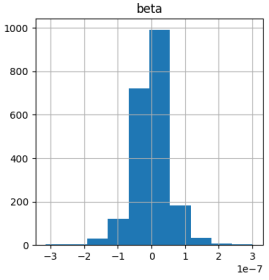



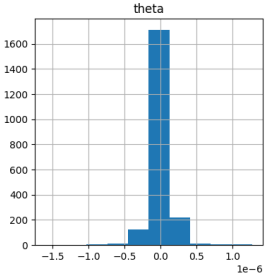



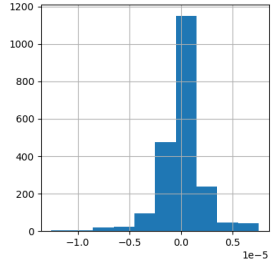

## sedate

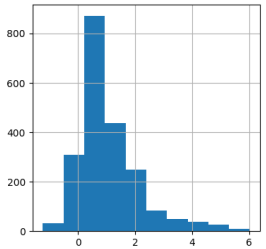


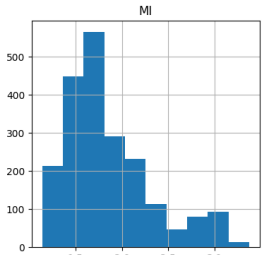



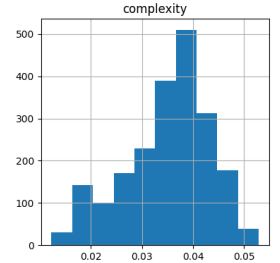



# ANALYSIS

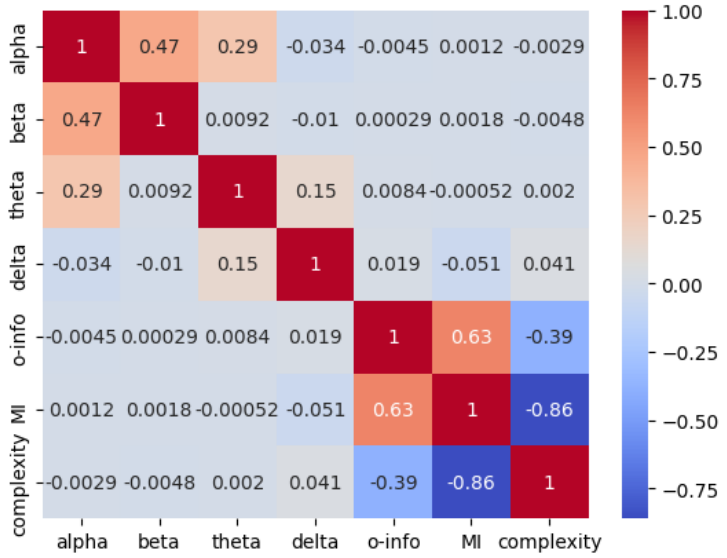
**Means**

| | O-info | MI | complexity |
|---|---|---|---|
| awake | 0.617820 | 1.572603 | 0.040230 |
| sedate | 1.052170 | 1.852189 | 0.035071 |

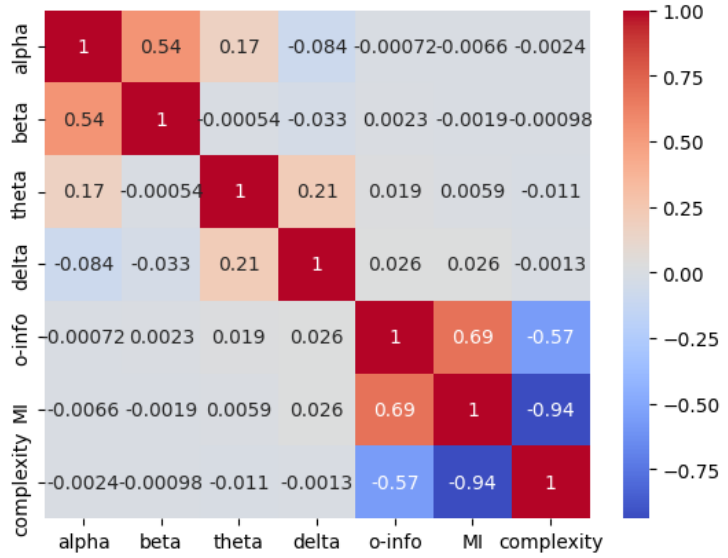
**T-test**

| | O-info | MI | complexity |
|---|---|---|---|
| T-statistic | -15.9771 | -24.2468 | 21.5317 |
| P-value | 0.0000 | 0.0000 | 0.0000 |
| Result (alpha=0.05) | significant difference | significant difference | significant difference |

# ANALYSIS




awake



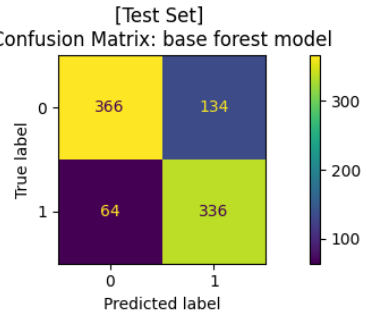

sedated

- IIT proxy are highly correlated especially MI and complexity

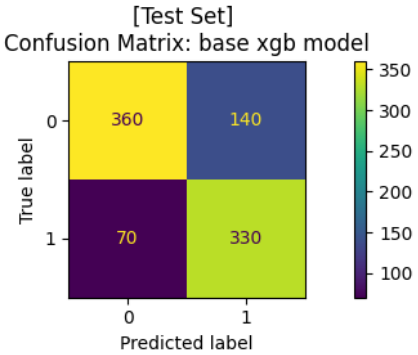
# CLASSIFICATION

- Spectral features only

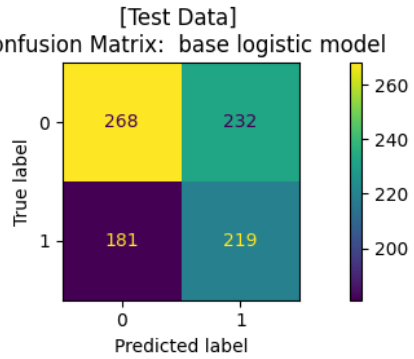
Random forest



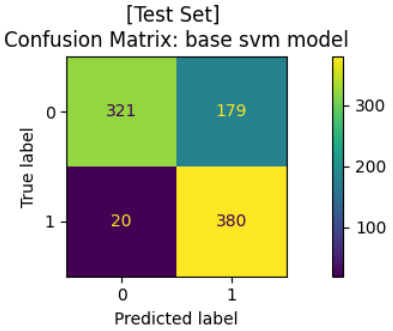

xgboost



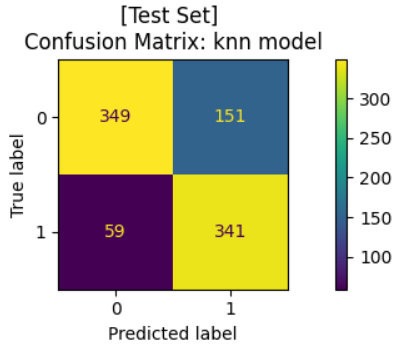

Logistic regression




svm




knn

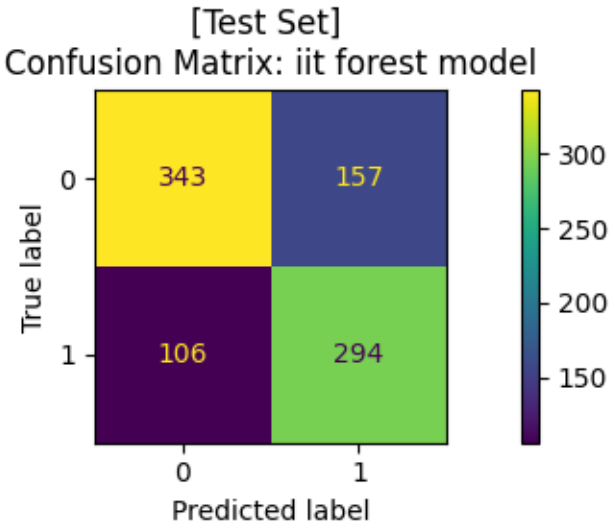


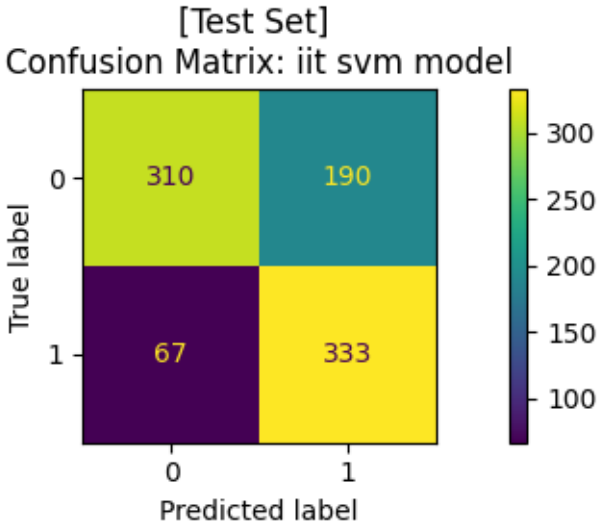

| Accuracy: | 78% | 77% | 54% | 78% | 77% |
|---|---|---|---|---|---|

Since SVM and Random Forest yield the best results, only they are selected to train the other feature sets

# CLASSIFICATION

- Iit features only



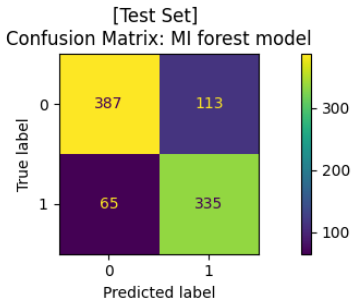



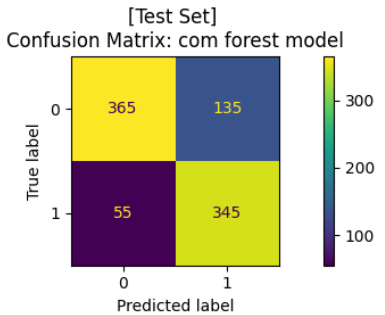

Accuracy: 71%

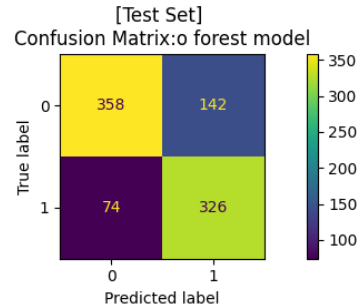
71%

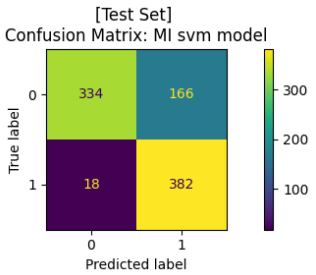
# CLASSIFICATION

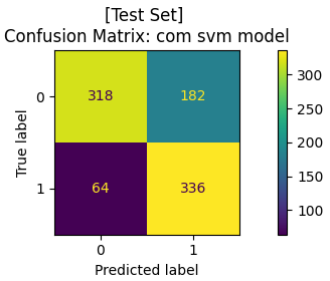
| Feature Set | Accuracy |
|---|---|
| Spectral + MI | 80% |
| Spectral + Complexity | 79% |
| Spectral + O-information | 78% |
| Spectral + All IIT | 79% |

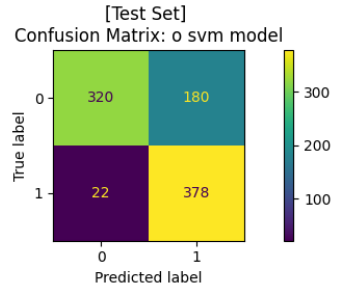


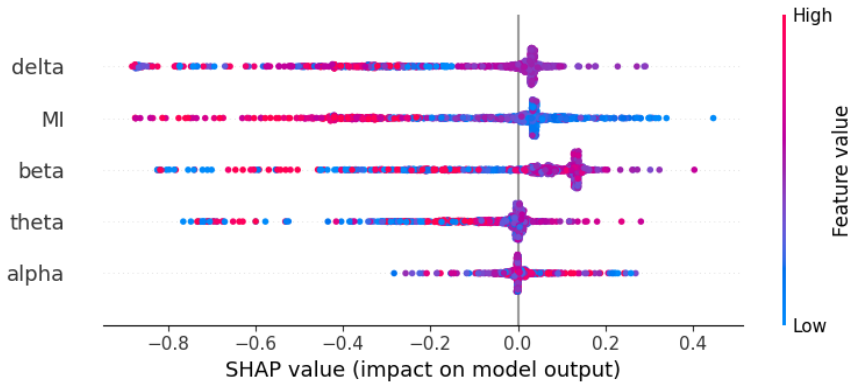



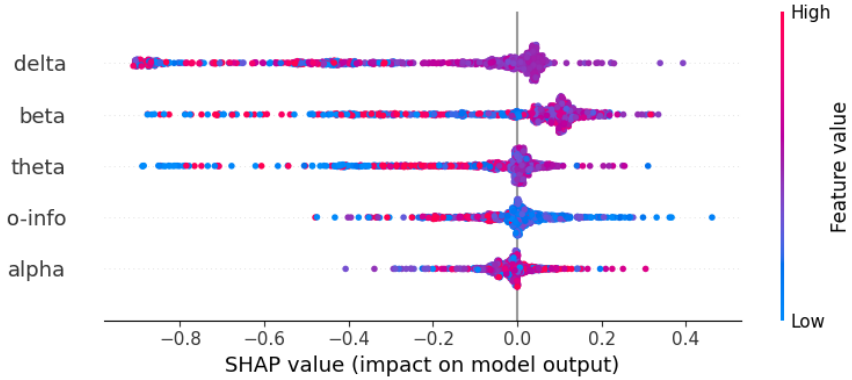



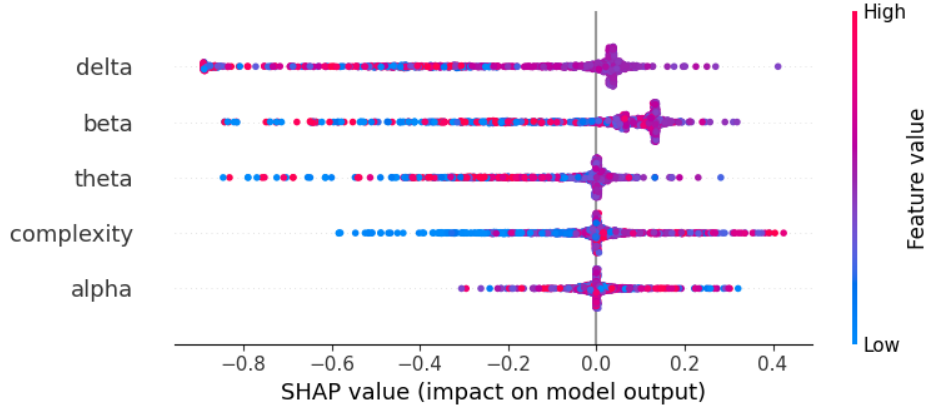



# CLASSIFICATION




Shap results:

- lower MI pushes model to predict awake state
- Higher complexity pushes model to predict awake state
- Lower o-info pushes model to predict awake state
- The results agree with the t-tests

# FINDINGS

- Spectral EEG features provided the strongest standalone classification performance.
- All information-theoretic features showed statistically significant differences between awake and sedated conditions.
- Mean pairwise mutual information consistently improved classification when combined with spectral features.
- Lempel–Ziv complexity decreased during sedation, consistent with previous anesthesia studies.
- Pairwise mutual information increased during sedation, suggesting stronger statistical dependence between EEG channels.
- O-information also increased during sedation, indicating a possible shift toward more redundancy-dominated interactions.

# INTERPRETATION

- Integrated Information Theory proposes that conscious states are characterized by richer and more integrated information processing. Since direct computation of integrated information (Φ) is computationally infeasible for EEG, this study evaluated several information-theoretic proxy measures.
- The results indicate that these proxies are sensitive to changes between awake and propofol-sedated EEG. Among the evaluated measures, mean pairwise mutual information provided the greatest complementary information beyond conventional spectral features, whereas complexity and O-information showed significant state differences but limited improvement in classification performance.
- These findings are consistent with the broader literature reporting reduced EEG complexity during anesthesia and increased statistical dependence among cortical signals during propofol sedation. However, because the evaluated measures are proxies rather than direct estimates of Φ, the results should be interpreted as being consistent with, rather than direct validation of, Integrated Information Theory.

# LIMITATION&FUTURE WORK

- Limitations
  - Pairwise mutual information is not equivalent to integrated information (Φ).
  - O-information remains an emerging measure with limited validation in anesthesia.
  - Results are specific to propofol sedation.
  - Feature extraction was performed on fixed EEG windows.
- Future work
  - Deep learning on raw EEG
  - Functional connectivity networks
  - Exact Φ approximations

# CONCLUSION

- This project investigates whether information-theoretic EEG features inspired by Integrated Information Theory provide complementary information to conventional spectral biomarkers for distinguishing awake and propofol-sedated states. Using subject-wise machine learning and SHAP analysis on the OpenNeuro ds005620 dataset, the study found that while spectral features remained the strongest standalone predictors, mean pairwise mutual information consistently improved classification performance when combined with spectral features. Lempel–Ziv complexity decreased under sedation, in agreement with established anesthesia literature, whereas increased pairwise mutual information suggested stronger inter-channel statistical dependence during propofol sedation. Overall, the findings indicate that IIT-inspired proxy measures are sensitive to changes in brain state, with pairwise mutual information emerging as the most informative complementary feature among those evaluated, while also highlighting the distinction between these practical EEG-derived proxies and the theoretical integrated information measure (Φ) proposed by IIT.

# REFERENCE


- https://en.wikipedia.org/wiki/Integrated_information_theory
- https://pmc.ncbi.nlm.nih.gov/articles/PMC9749579/
- https://www.bitbrain.com/blog/qeeg-brain-mapping
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